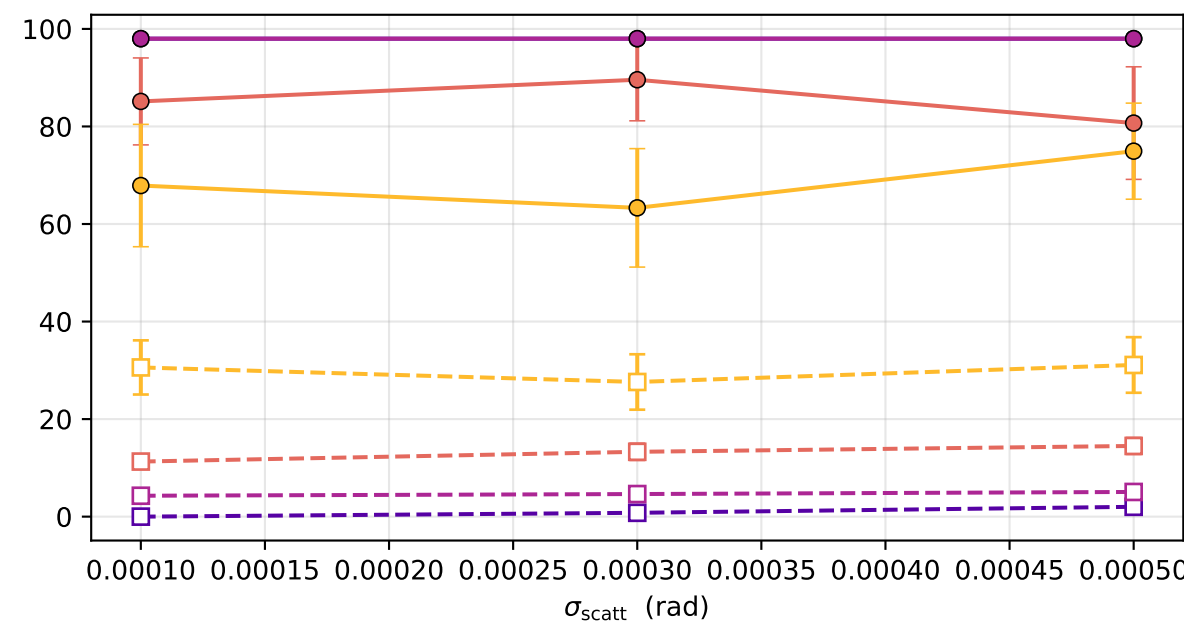
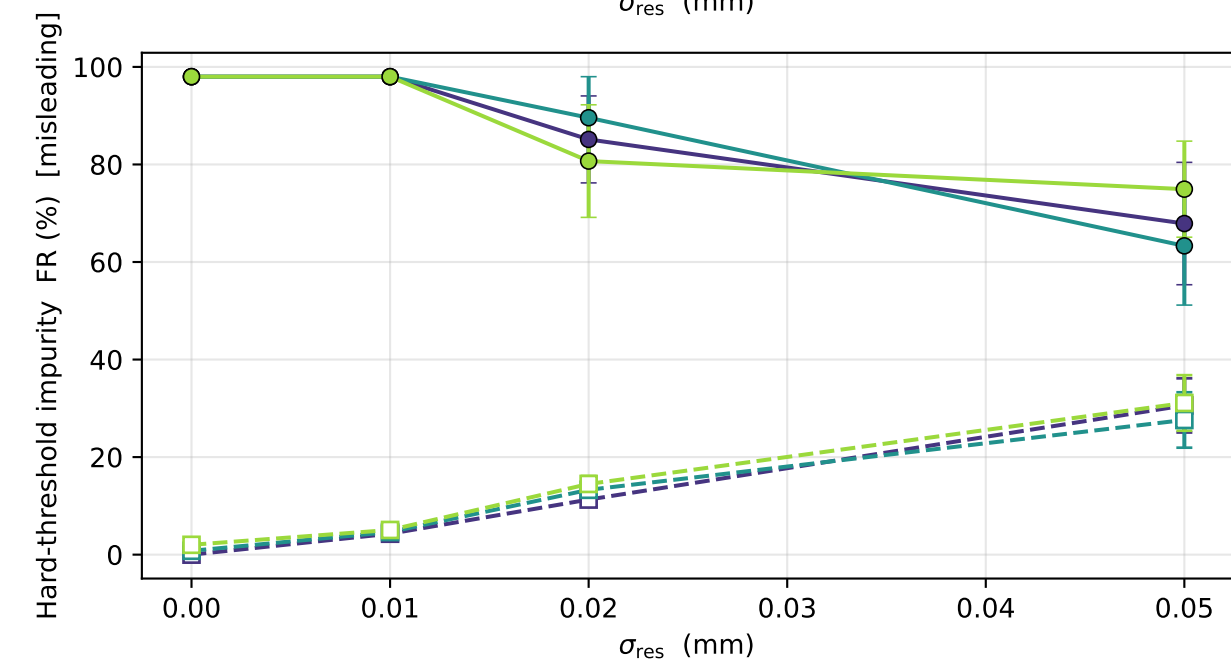
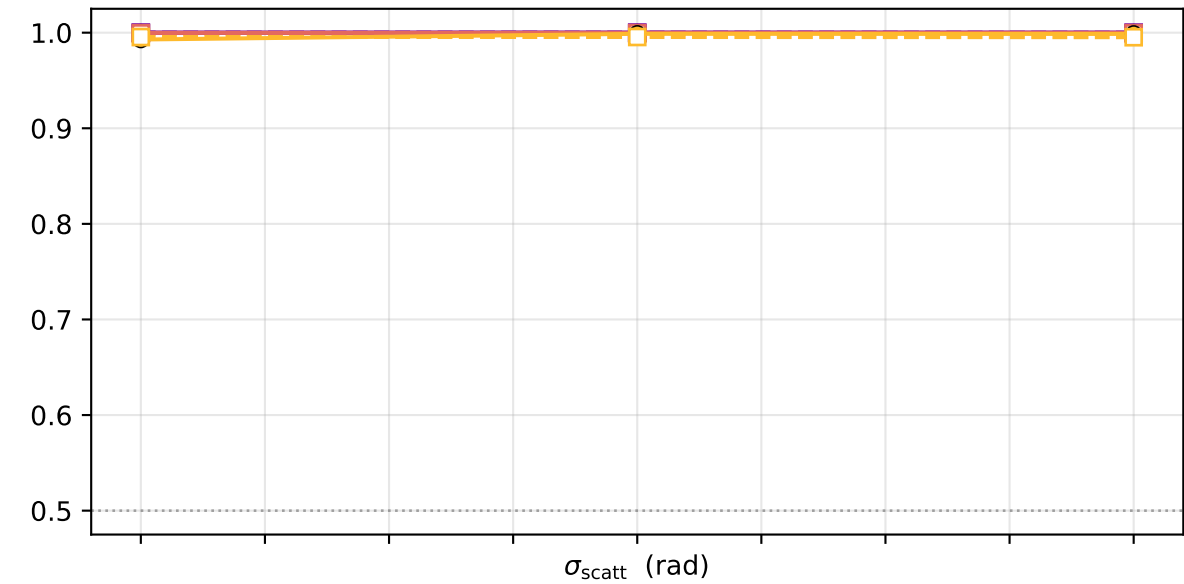
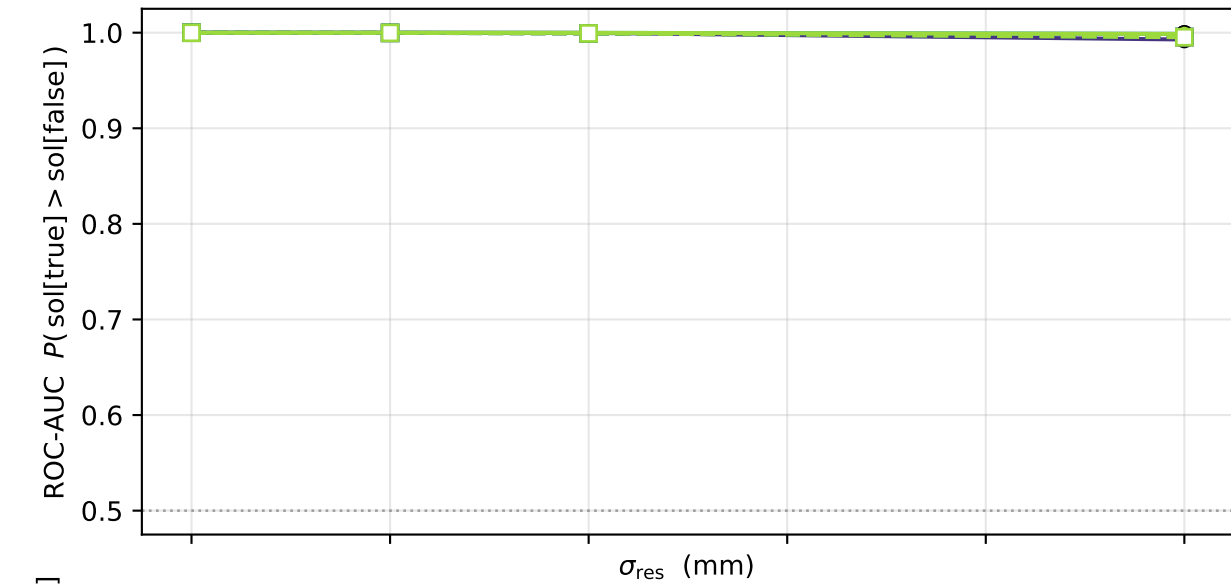
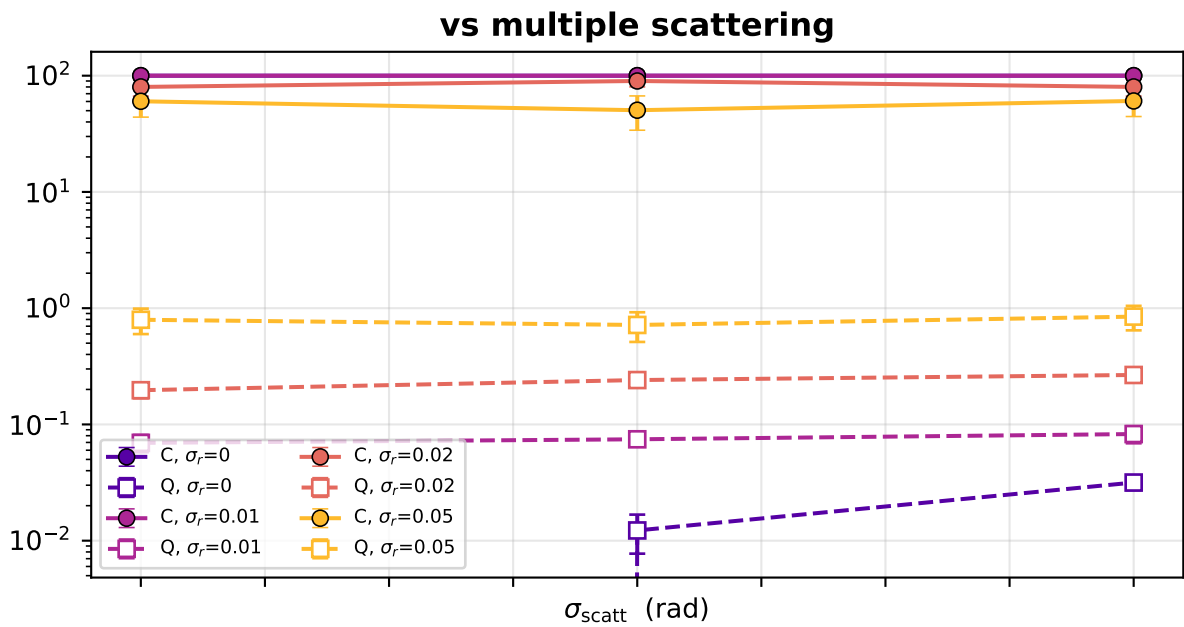
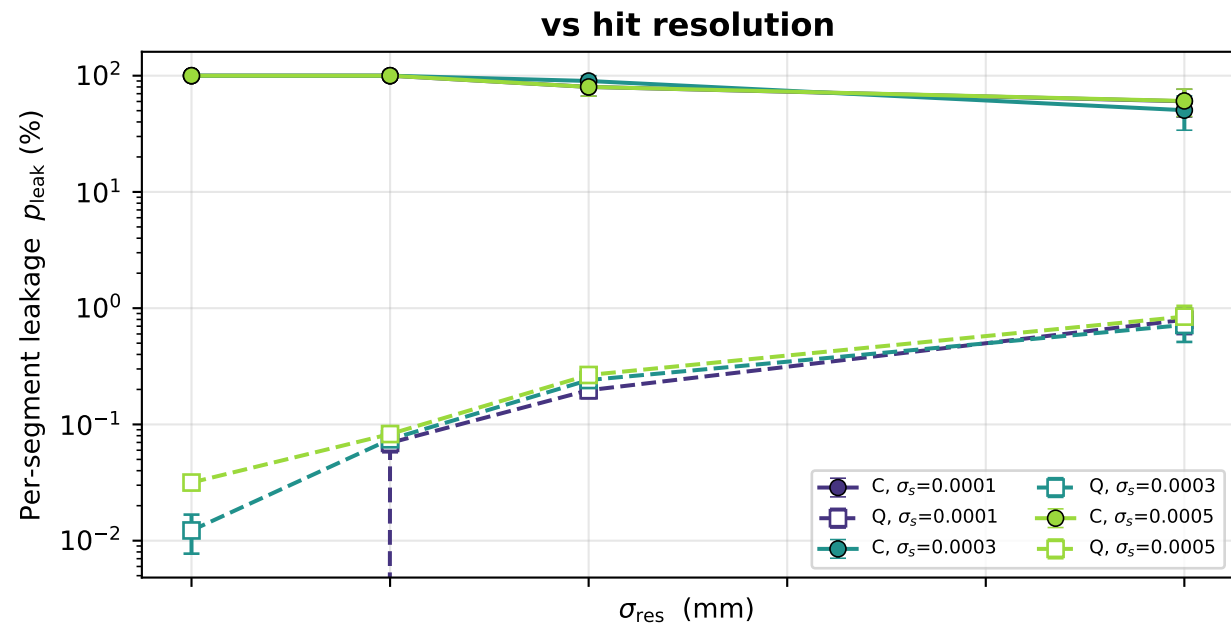


Threshold-robust segment metrics — $T = 50$ tracks (solid = classical, dashed = 1-bit-HHL)



Controls: $\gamma=3, \delta=1, \phi_{\text{max}}=\theta_{\text{max}}=0.2 \text{ rad}, \theta_{\text{min}}=1.5\text{e-}5, \tau=0.35, \varepsilon=\text{calc.}(s=3), \text{conv OFF}, \text{hit-ineff } 0, 5 \text{ planes } \Delta z=33 \text{ mm}, \text{statevector } 1\text{BQF}$
 Fixed here: per-segment leakage & impurity at $\tau=0.35$; AUC threshold-free
 Swept: $\sigma_{\text{res}}, \sigma_{\text{scatt}}, T$